

PHASE 4 UPDATE FOR ADITYA SIR

How long does a vision-language model need to look at the image?

It depends less on what the task is about and more on one thing: whether the model has to write the answer or pick it from options.

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THE TEST

Take the image away after layer L, and see what the model still gets right



- Every visual token is deleted after layer L. The question keeps going up the network without the picture.
- Layer 0 means no image at all. That gives the floor: what the model scores from the question alone.
- Run for every L, on 8 tasks and 3 models: Qwen2.5-VL 7B, Qwen2.5-VL 3B and InternVL2 2B.

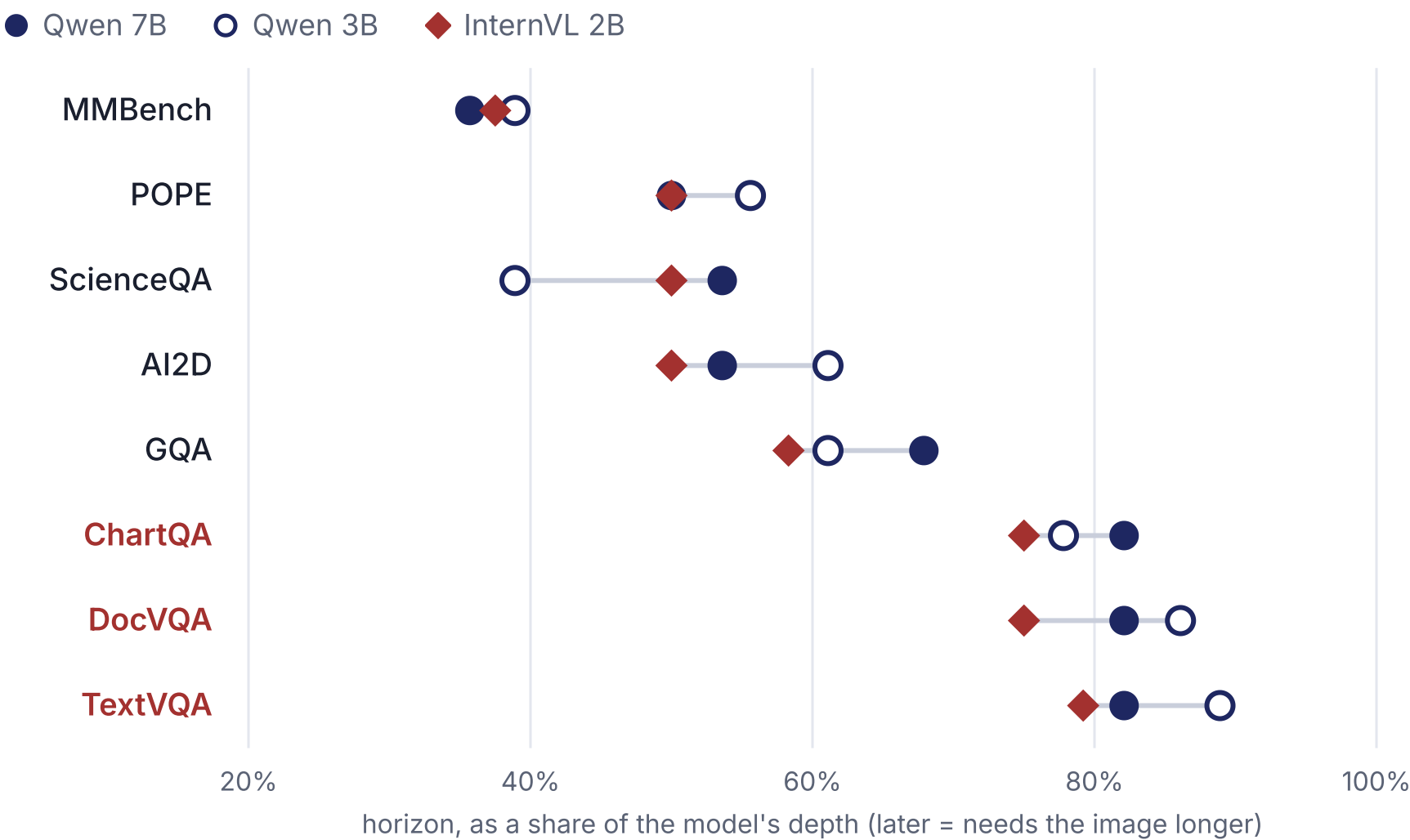
How we read one curve (Qwen 7B, MMBench)

With the image: **88**. Without it: **40**. So the image is worth **48** points.

The **horizon** is the layer after which the model holds on to at least half of those 48 points for good. Here that is layer **10** of 28.

FINDING 1

Tasks need the image for different lengths of time, and the order holds across models



Do the three models rank the 8 tasks the same way?

Pair	Rank agreement
Qwen 7B and Qwen 3B	0.90
Qwen 7B and InternVL	0.96
Qwen 3B and InternVL	0.93

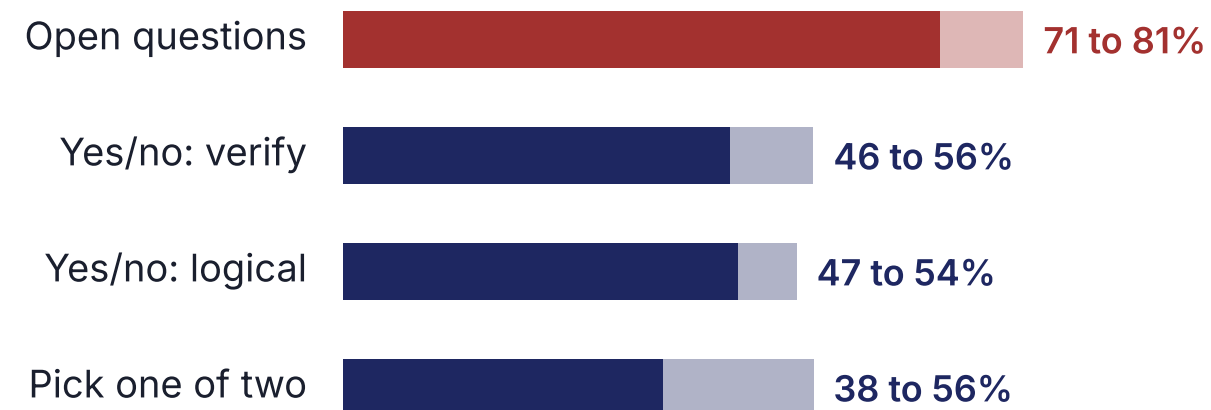
Spearman correlation, all $p < 0.01$. The two cross-family pairs agree as well as the two Qwen models do.

The **order** carries over between models. The exact layer does not, so it still has to be measured per model.

THE CATCH

Every early task was multiple choice. Every late task made the model write its answer.

So "reading needs the image longer" and "writing the answer needs the image longer" were tangled together. GQA was the first hint, because it mixes both kinds of question:



share of depth; bar spans the 3 models; same 300 questions

"What brand is on the bottle?"

OPEN-ENDED

The model has to spell out "Coca-Cola". It needs the actual letters, so it keeps looking until late.

MULTIPLE CHOICE

A Pepsi

B Coca-Cola

C Fanta

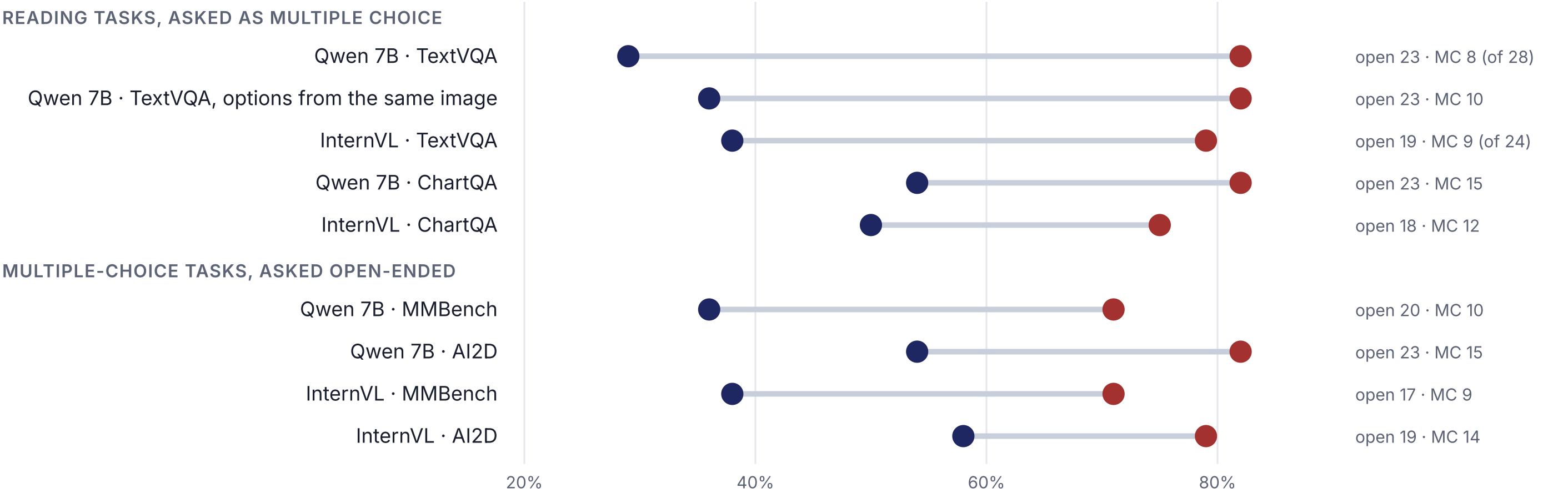
D Sprite

A glance at the red logo is enough to pick B.

FINDING 2

Change only the answer format, and the horizon moves. In both directions.

● open-ended ● multiple choice Same questions and images in both formats. Horizon shown as a share of the model's depth.

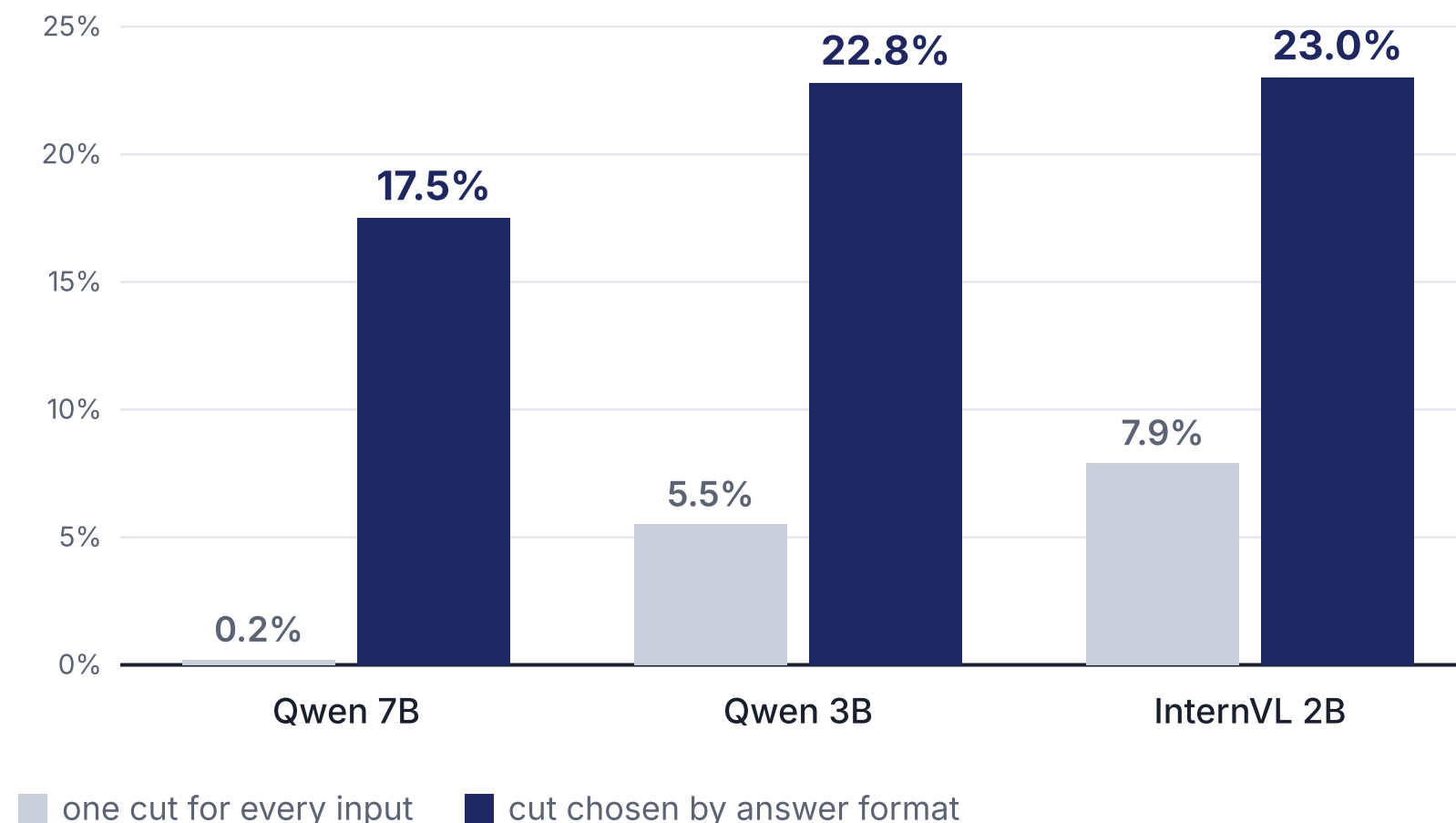


Writing an answer keeps the model looking almost to the top. Picking from options is settled by the middle. Every shift's paired 95% interval excludes zero.

WHAT IT MEANS FOR PRUNING

A pruner can read the format from the prompt, and drop the image early only when that is safe

Share of layers that no longer carry visual tokens, with every task keeping at least 98% of its accuracy



- Without the format, one cut has to protect the reading tasks, so it barely moves from the top.
- With it, Qwen 7B keeps the image for all 28 layers on open questions and drops it after layer 20 on closed ones.
- For comparison, the released BTP deletion saves 21% on Qwen 7B but leaves the worst task at **22%** of its accuracy.
- These numbers come straight from stored runs, which record right or wrong for each question at each layer. Cuts learned on the 8 tasks also held on the format-control sets they never saw.

CHECKED ON GPU

Inside BTP: keep the final deletion where it is safe, remove it where it is not

Qwen2.5-VL 7B	Baseline	Released BTP	Corrected BTP	Format-aware
CLOSED-FORM: IMAGE DROPPED AFTER LAYER 20				
POPE	87.6	86.2	86.3	86.3
MME (perception)	1674.5	1658.7	1651.2	1664.1
MMBench	83.7	79.3	79.0	79.6
ScienceQA	88.1	85.1	85.0	85.1
AI2D	86.4	79.6	79.2	79.6
OPEN-ENDED: NEVER DROPPED				
TextVQA	86.2	23.3	80.5	80.5
DocVQA	94.7	19.2	76.8	76.8
ChartQA	76.8	29.0	45.4	45.4

- Closed-form tasks lose nothing when the image goes two layers earlier than the released code drops it. All five match or beat both other versions.
- Reading tasks recover from 28% to 78% of baseline.
- One caveat: inside BTP the extra compute saved is small, because only 12.5% of visual tokens are left after layer 16. The bigger saving is on slide 6.
- ChartQA stays low in every version. There, the pruning itself costs more than the deletion.

What I think we can claim, and where I would like your view

The claim

How long a vision-language model needs its visual tokens depends mainly on the answer format. The ordering of tasks is stable across model sizes and families, and a pruner that knows the format saves 17 to 23% of visual layers at 98% accuracy, where a single cut saves under 8%.

What could weaken it

- Three models, two families, 200 to 300 questions per task.
- Reading questions turned into multiple choice still score 0.51 to 0.61 with no image, so the options alone rule some answers out.
- Open-ended MMBench and AI2D score only 0.28 to 0.40, so those curves are noisier.
- Content still matters a little: ChartQA keeps about a third of its gap even as multiple choice.

Questions for you

1. For the paper, should the framing be "answer format" rather than "task families"?
2. Next step: detect the format from the prompt inside the model, and add a larger model or a third family?
3. Is this ready to take to ma'am, or should we close any of the gaps on the left first?

Code, raw results and all numbers: github.com/milab-iitj-dev/btp-reproduction, folder phase4-access-horizon.